

Luxury-scoring model comparison — plain-language guide

A companion to [report.md](#). The report is the raw numbers; this document explains what each number means, how to read every table, what this particular run is telling us, and what to do about it. Where the report shows a table, this guide tells you which direction is "good" and what the current values imply.

1. What this experiment actually tested

The production pipeline scores each listing photo for how *luxurious* the room looks, on a per-room scale (kitchens 1–7; bathrooms, bedrooms, living rooms 1–8). Today that scoring is done by **gpt-4o**. The question is: **is there a better or cheaper model we should switch to?**

To answer that we took **152 real listing images** (~38 per room type, at most one image per room type per property so no single home dominates), and for each image we asked **five models** to score it **five times each**, using the exact same rubric prompt and anchor-grid reference image the production `score.py` uses. Five repeats per image is what lets us measure whether a model is *consistent* (does it give the same answer when asked the same question?).

Separately, **three humans** (harvey, robin, seb) scored all 152 images by hand. That gives us a human ground-truth to compare the models against — not just "which model agrees with gpt-4o" but "which model agrees with people."

The five models:

model	who makes it	role in this test
gpt-4o	OpenAI	the incumbent — what we run today
gemini-2.5-flash	Google	cheap challenger
gemini-3.5-flash	Google	newer, pricier challenger
claude-haiku-4-5	Anthropic	cheap challenger
claude-sonnet-4-6	Anthropic	mid-tier challenger

2. Consistency — "does the model repeat itself?"

Report section: *Consistency (within-image std vs between-image std)*

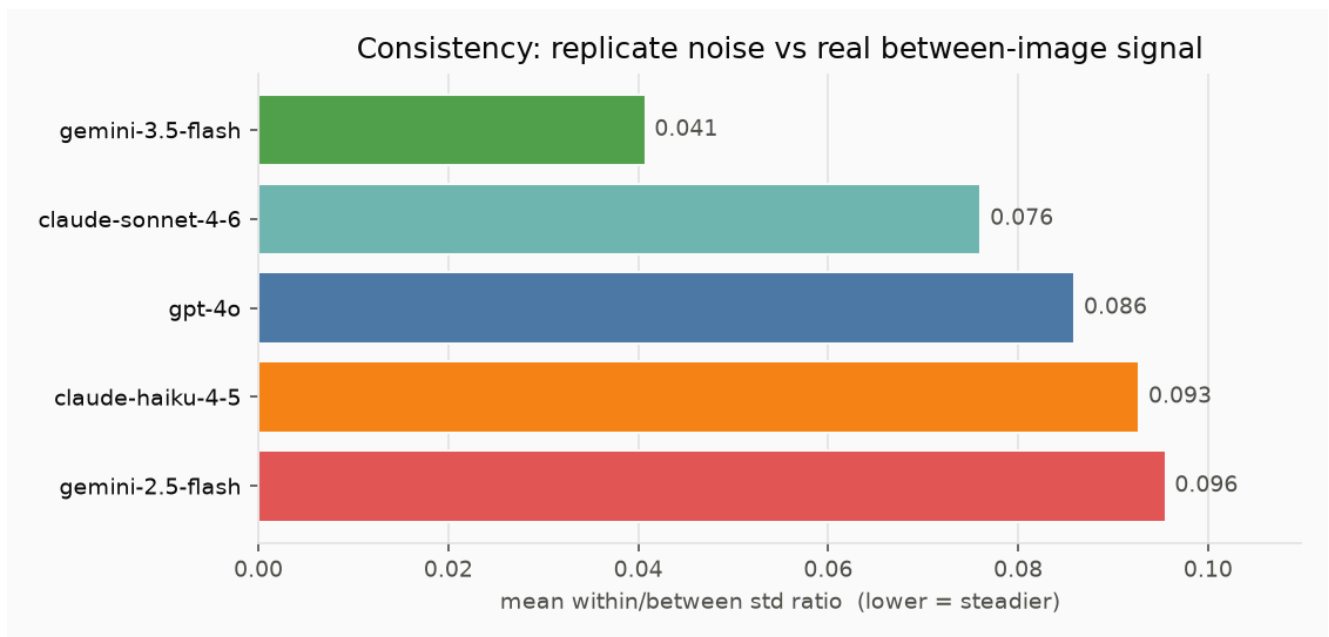
This is the single most important quality metric, and here's the intuition. Every image was scored 5 times. Two kinds of variation exist:

- **within-image std** — how much a model's *own five scores* for the *same image* jump around. This is pure noise. You want it near zero: ask the same question, get the same answer.
- **between-image std** — how much the average scores differ *across different images*. This is real signal — luxurious rooms should score higher than plain ones, so this should be large.

The headline number is the **ratio = within ÷ between**. It answers: *is the model's random noise small compared to the real differences it's supposed to be detecting?*

- **ratio well below 1 = good**. Noise is tiny next to signal.

- **ratio near or above 1 = bad.** The model's own randomness is as big as the differences between properties, so its scores aren't trustworthy.



What this run shows (mean ratio across room types):

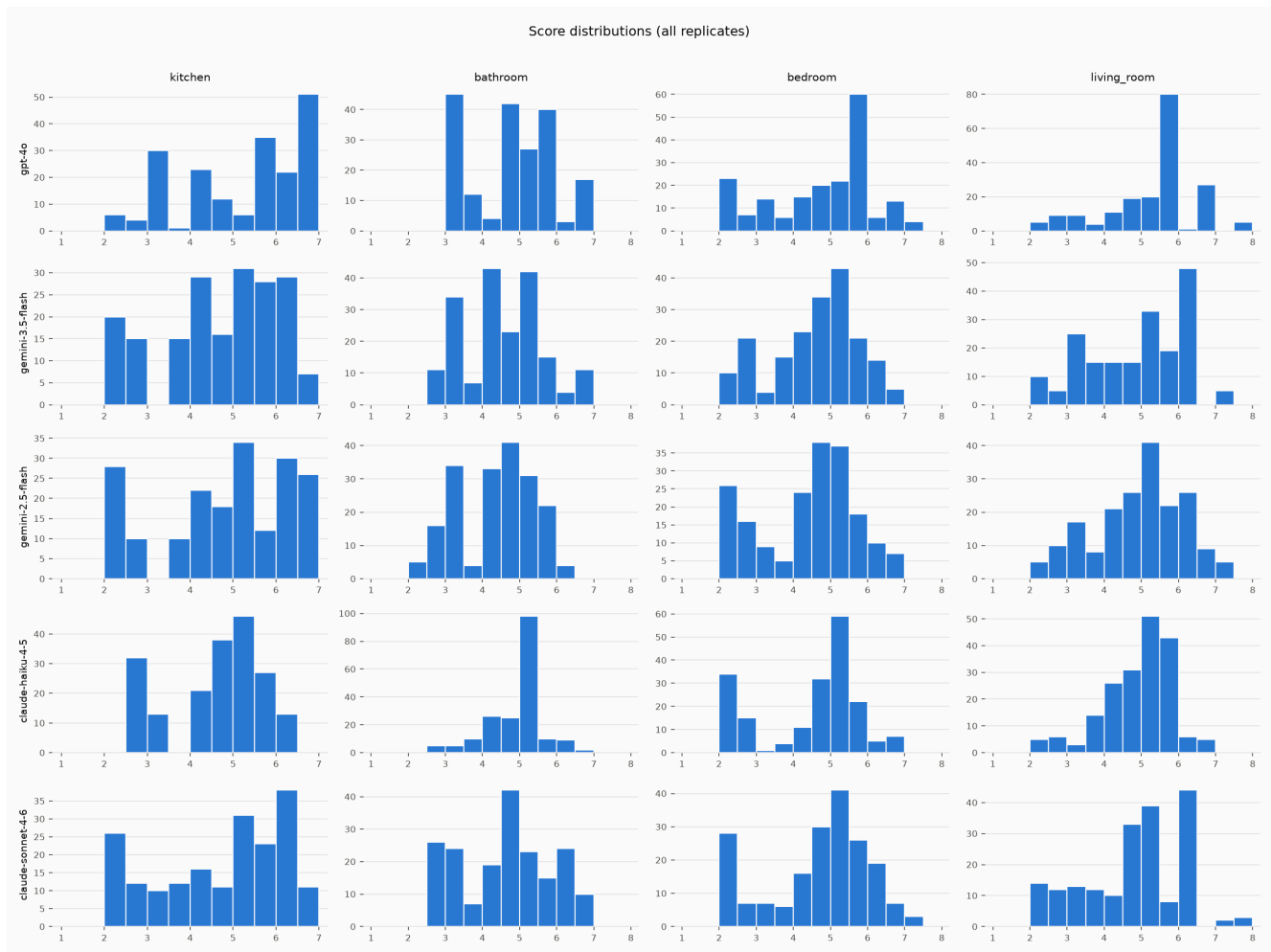
model	mean ratio	read
gemini-3.5-flash	0.041	far and away the steadiest — its repeats barely move
claude-sonnet-4-6	0.076	very stable
gpt-4o	0.086	the incumbent; solid
claude-haiku-4-5	0.093	fine
gemini-2.5-flash	0.096	still fine, but the noisiest of the five

All five are well under 1, so **every model is usable** — none is just guessing. But gemini-3.5-flash is in a class of its own: roughly **2x steadier** than gpt-4o and the rest. If you re-score the same photo tomorrow, it's the least likely to give you a different number.

3. Score distributions — "does the model use the whole scale?"

Report section: *Score distributions* (histograms + compression flags)

A model can be consistent but still useless if it rates almost everything a 5. These histograms show the spread of scores each model gives per room type.



The **compressed flag** is a warning light. It trips when a model either barely touches the top/bottom of the scale, or uses less than 60% of the available range. `compressed = True` means "this model bunches its scores in the middle and rarely commits to *this is a dump* or *this is stunning*."

How to use it: don't treat `compressed = True` as disqualifying — it's normal for real listing photos to cluster in the middle (most rooms are average). Use it as a tie-breaker. A model with less compression is drawing sharper distinctions between properties. In this run the flag trips for most model/room combinations (bathrooms especially — real bathrooms don't vary much), which is expected. Kitchens show the most spread across all models.

4. Agreement with gpt-4o — "how different is each challenger from today?"

Report section: *Spearman rank correlation of mean scores vs gpt-4o*

Spearman correlation measures whether two scorers *rank* the images the same way — do they agree on which rooms are fancier than which? It ignores absolute offsets (one scorer running 0.5 higher across the board doesn't hurt it); it only cares about order. Runs from -1 (opposite) through 0 (unrelated) to **1 (identical ranking)**.

This table is **vs gpt-4o**, so it answers "if we switched, how much would the ordering change from what we ship today?" — *not* "which is more correct." gpt-4o is the baseline, not the truth.

What this run shows (overall): gemini-3.5-flash 0.938, gemini-2.5-flash 0.926, claude-sonnet-4-6 0.921, claude-haiku-4-5 0.818. So the two Geminis and Sonnet are all **very close to gpt-4o** (0.92–0.94) — switching to any of them would mostly preserve today's ordering. claude-haiku-4-5 drifts the most (0.82): it's the biggest behavior change from the incumbent.

Disagreements vs gpt-4o (|mean diff| ≥ 2 levels): None. No model ever landed a full 2 scale-points away from gpt-4o on any image. There are no blow-up cases to inspect — differences are gradual, not wild.

5. Room-type sanity check

Report section: *Room-type disagreement rate*

During scoring, each model also states what room it thinks it's looking at. This table checks that against the Stage 1 label. **Every value is 0** — the models never disagreed about what room a photo showed. This is purely a data-integrity check (are we scoring kitchens with the kitchen rubric?), and it passed cleanly. Nothing to act on.

6. Cost — "what does a full production run cost?"

Report section: *Cost*

`mean_cost_per_call_usd` is measured directly from the API responses (real token counts, real prices, with prompt caching already applied). The `full_run_*` columns extrapolate to scoring the **entire corpus** — an estimated **41,779 scoreable images** — once each. `batch` = using each provider's async batch API, which is ~half price but slower (fine for a bulk backfill, not for live scoring).

What a full run costs (batch price):

model	per call	full run (batch)	full run (standard)
gemini-2.5-flash	\$0.0030	\$63	\$125
claude-haiku-4-5	\$0.0046	\$97	\$194
gpt-4o (today)	\$0.0049	\$101	\$203
claude-sonnet-4-6	\$0.0058	\$122	\$243
gemini-3.5-flash	\$0.0105	\$220	\$439

The spread is ~3.5× from cheapest to most expensive. gemini-3.5-flash — the quality winner — is also **the most expensive**, about 2× the cost of running gpt-4o today. gemini-2.5-flash is the cheapest by a wide margin. Latency (also in the table) ranges ~4.5–9s per call and matters only if you score live rather than in batch.

7. Agreement with humans — the most important section

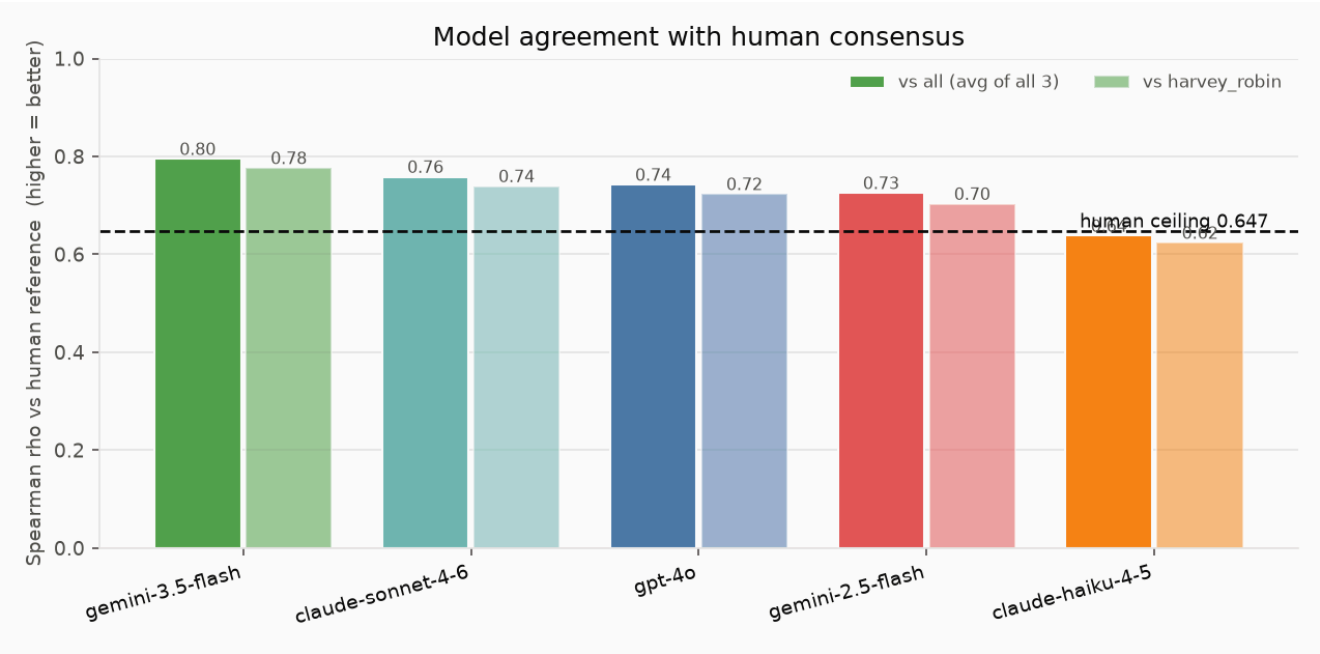
Report section: *Human agreement (Stage 2b ratings)*

Sections 4–5 compared models to *gpt-4o*, which is circular — gpt-4o isn't necessarily right. This section compares every model to **actual human judgment**, which is the real target. Three people scored all 152 images, and we compare each model against several **human references**:

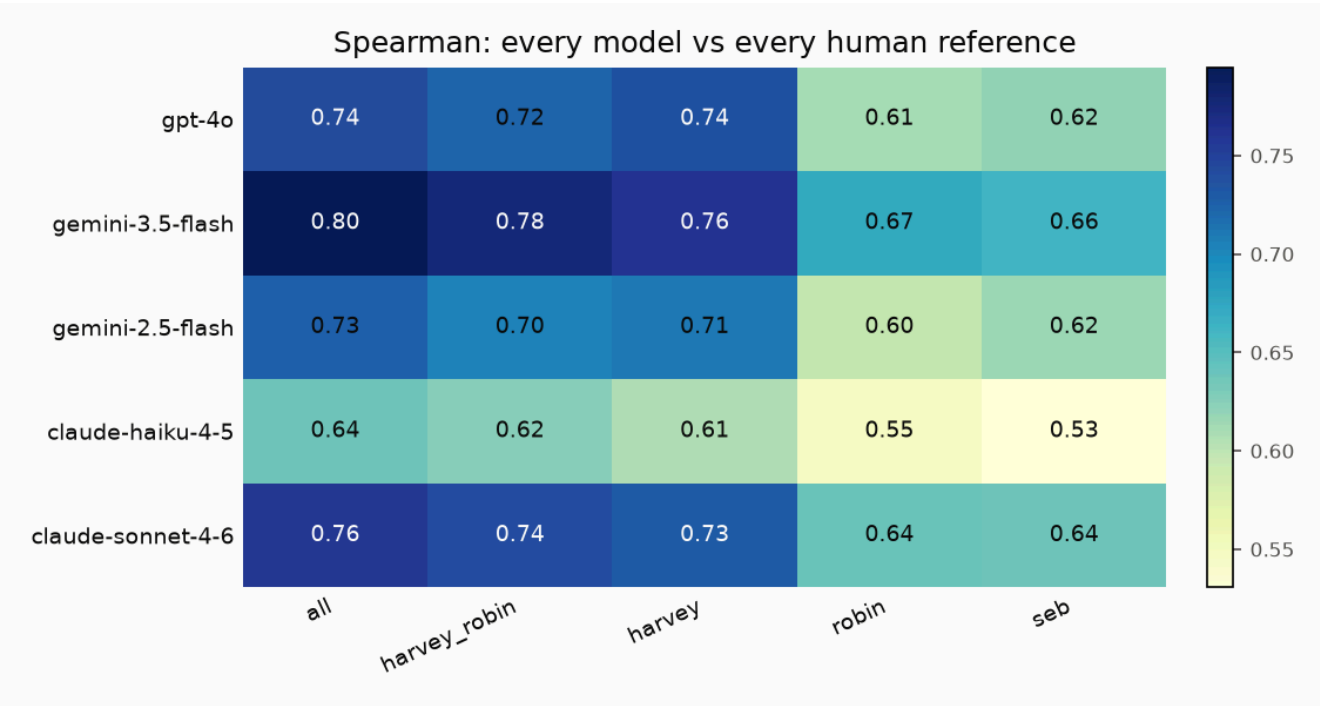
- `all` — the average of all three raters (the consensus "house view").
- `harvey_robin` — the average of just harvey and robin.
- `harvey`, `robin`, `seb` — each person on their own.

Averaging raters cancels individual noise, so the group references (`all`, `harvey_robin`) are the most stable targets; the individual columns show how each person's personal taste lines up with the models.

There are **three tables**, each answering a different question:



The dashed line is the human ceiling (0.647). Bars above it track the group consensus at least as well as an individual human would — see 7d for why that's possible. The full picture, including each rater individually, is the heatmap below (darker = stronger agreement):

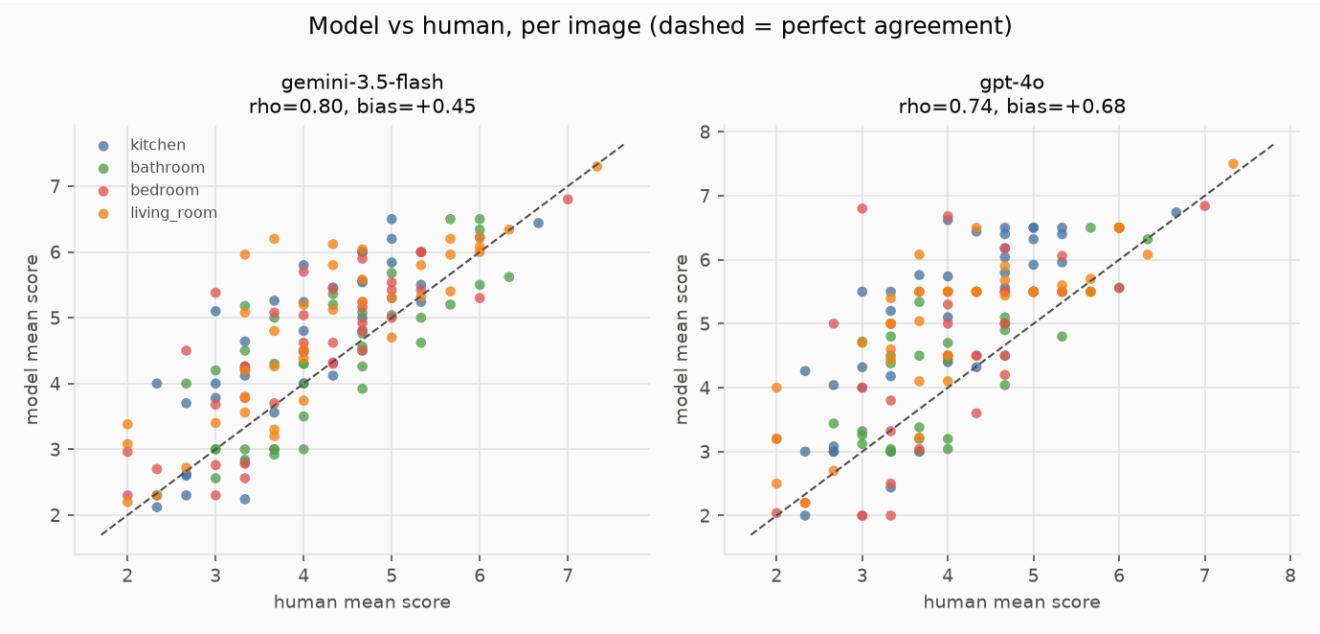


7a. Spearman rho — do model and human *rank* rooms the same? (higher = better)

model	vs all	vs harvey_robin	read
gemini-3.5-flash	0.795	0.776	best human agreement of any model
claude-sonnet-4-6	0.758	0.739	second
gpt-4o (today)	0.742	0.723	the incumbent
gemini-2.5-flash	0.725	0.703	slightly below gpt-4o
claude-haiku-4-5	0.638	0.624	clearly the weakest vs humans

gemini-3.5-flash agrees with people best, and by a real margin over gpt-4o. claude-haiku-4-5 is the one to avoid on quality grounds.

Seeing it per-image makes the correlation and the bias concrete — the winner sits tighter to the diagonal, and both models sit mostly *above* it (scoring higher than people):



7b. Mean absolute deviation — how many *points* off is the model, typically? (lower = better)

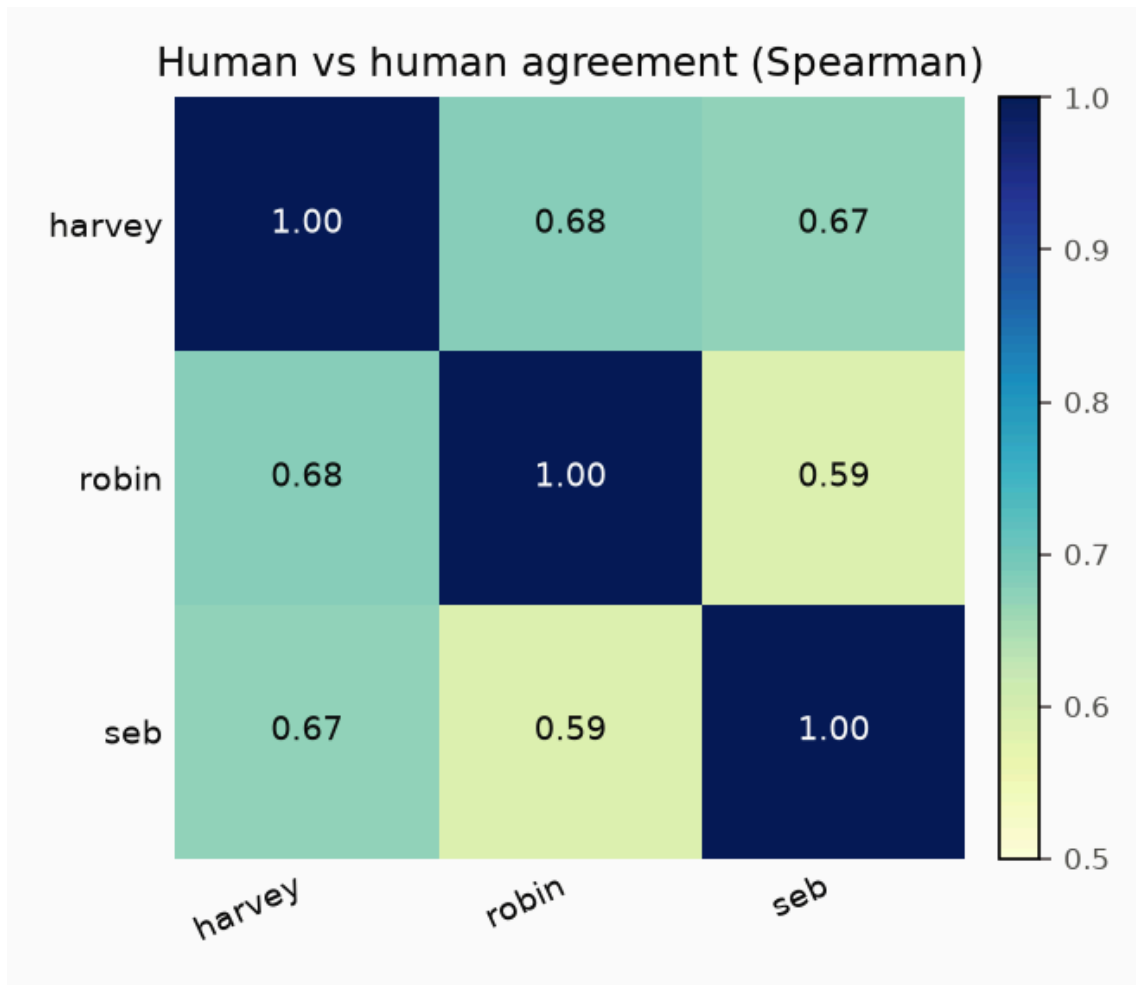
This is in actual score units. 0.67 means the model's average score for an image is, on average, two-thirds of a point away from the humans'. Against all, gemini-3.5-flash is closest (0.67) and gpt-4o is farthest (0.90). Note every model is *further* from seb ($\approx 1.0+$) than from harvey — see 7d.

7c. Signed bias — does the model run *high* or *low* vs humans? (near 0 = neutral)

Positive means the model scores **higher** than the human. **Every model has a positive bias against every rater** — the models are uniformly more generous than the people, gpt-4o most of all (+0.68 vs all). This is easy to correct (you can subtract a constant), and because it's a roughly uniform offset it doesn't hurt the ranking metrics in 7a. Worth knowing if the absolute score value is shown to users or fed into a threshold.

7d. Humans vs humans — the reality check (inter-rater matrix + ceiling)

Before judging the models, ask: *how well do the humans even agree with each other?* The pairwise Spearman among raters: harvey-robin 0.68, harvey-seb 0.67, robin-seb 0.59. The **human-agreement ceiling is 0.647** — the average person-to-person agreement.



This ceiling is the yardstick. **No model should be expected to agree with the average human more than humans agree with each other**, because the humans themselves don't have a single "right" answer — luxury is subjective. And indeed the top models (gemini-3.5-flash at 0.795, Sonnet at 0.758, gpt-4o at 0.742) score *above* the 0.647 ceiling against the `all` reference. That's not magic — it's because the `all` reference is an *average* of three people, and an average is smoother and more predictable than any single noisy rater. It means these models track the group consensus at least as well as an individual human would.

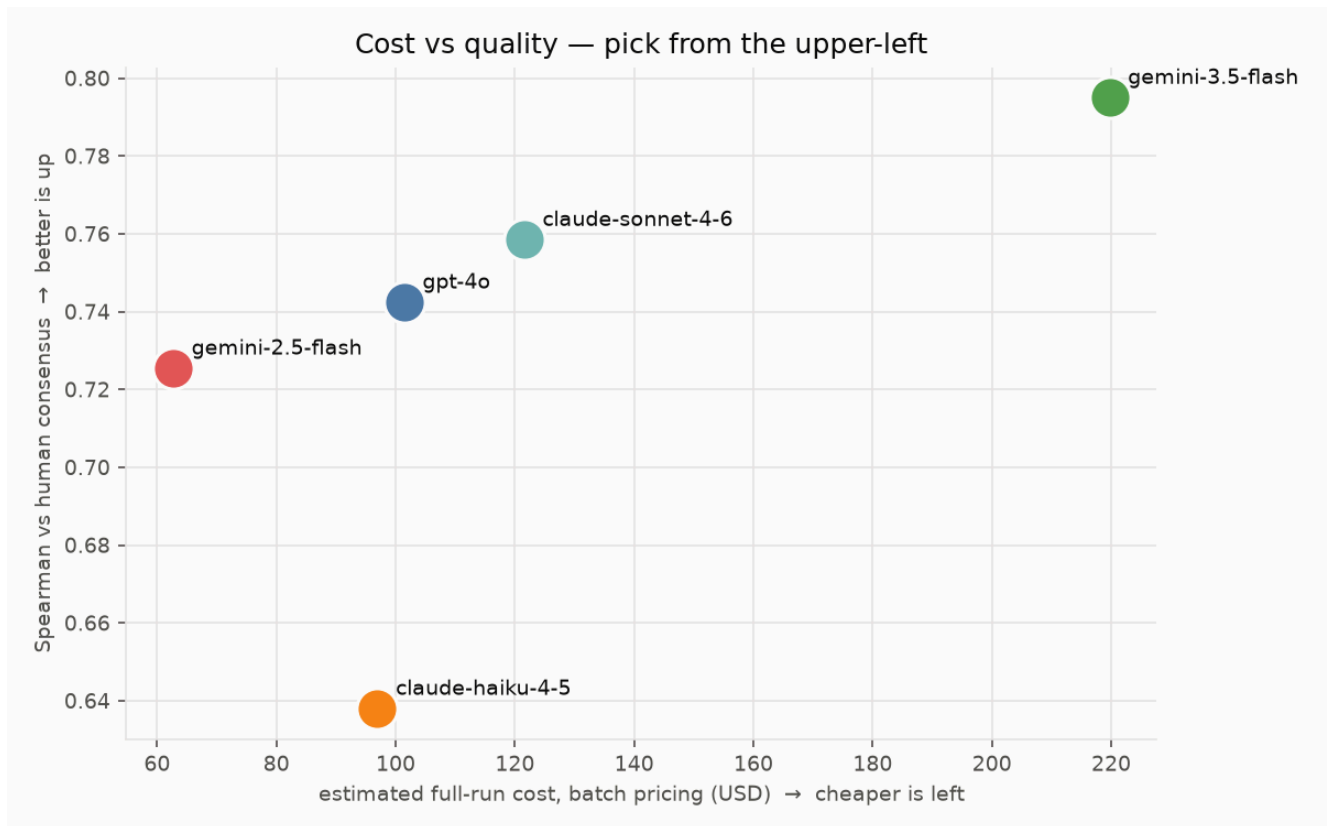
On the raters: none is flagged as an outlier (all agree with the others about equally, 0.63–0.68). seb is very slightly the odd one out — every model is harsher-looking against seb because **seb rates lower than harvey and robin**, so the models' generous scores sit farther above his. That's a difference in seb's personal scale, not a data problem. harvey and robin are the tightest-agreeing pair, which is why `harvey_robin` is offered as a second consensus reference.

8. The recommendation

Report section: *Recommendation* → **Pick:** `gemini-3.5-flash`

How the pick is made: take the models that agree with the human consensus (`all`) within 0.05 of the ceiling, then among those prefer the most consistent, then the cheapest.

The whole trade-off on one chart — quality (up) against full-run cost (right). The ideal model is upper-left; gemini-3.5-flash buys the top quality at the top price, while claude-sonnet-4-6 sits on the efficient frontier between it and the incumbent:



gemini-3.5-flash wins on quality, cleanly:

- **Best human agreement** — 0.795 vs the human consensus, ahead of gpt-4o (0.742) and everything else.
- **By far the most consistent** — 0.041 ratio, ~2× steadier than gpt-4o; its repeat scores barely move.
- **Closest to humans in absolute points** (lowest MAD) and one of the smaller biases.

The catch is cost. It's the most expensive model tested — ~\$220 to score the full corpus in batch vs ~\$101 for gpt-4o today, roughly 2×.

So which should you actually pick?

It depends on what you're optimizing:

- **Best quality, cost is acceptable** → **gemini-3.5-flash**. It genuinely beats gpt-4o on both agreeing-with-humans and stability. If scoring cost isn't a pain point (a few hundred dollars for a full backfill, less for incremental new listings), this is the upgrade.
- **Best quality-per-dollar** → **claude-sonnet-4-6**. It's the *second* best on human agreement (0.758, above gpt-4o) and very consistent (0.076), for only a small cost premium over the incumbent (~\$122 vs ~\$101 full-run batch). You get most of gemini-3.5-flash's quality gain at nearly half its cost.
- **Cheapest** → **gemini-2.5-flash**. By far the lowest cost (~\$63 full run, ~40% cheaper than gpt-4o), but it's the *noisiest* model and lands slightly *below* gpt-4o on human agreement — you'd be trading a little quality for the savings.
- **Don't switch to claude-haiku-4-5** on quality — it's the weakest human agreement (0.638, below the incumbent) despite being cheap.
- **Staying on gpt-4o is defensible**. It's mid-pack on everything and there's no disagreement blow-up. The case to switch is that gemini-3.5-flash is *measurably* better with humans and much steadier — but it's not a correctness emergency.

Bottom line: if you'll pay for quality, **gemini-3.5-flash**. If you want the best quality-per-dollar without losing ground to today, **claude-sonnet-4-6**. The cheap option **gemini-2.5-flash** only makes sense if cost dominates and you can

tolerate its extra noise.

9. Caveats — how much to trust this

- **Sample size is 152 images, 3 raters.** The rankings are stable and the gaps between the top models and haiku are real, but small differences (e.g. gpt-4o vs gemini-2.5-flash) are within noise — don't over-read a 0.02 gap.
- **Luxury is subjective.** The 0.647 human ceiling proves even careful people disagree meaningfully. "Agrees with humans" means "agrees with *these three* humans' average," which is the best available proxy, not absolute truth.
- **The uniform positive bias** (models score higher than humans) is correctable with a constant offset and doesn't affect rankings, but matters if the raw score is user-facing or feeds a fixed threshold.
- **Costs are extrapolations** from measured per-call cost × an estimated 41,779 scoreable images; the real bill scales with how many images you actually score and whether you use batch pricing.
- **All numbers regenerate** from `stage5_report.py` . If you collect more ratings or re-score, re-run it and this guide's specific figures may shift (the way to read them won't).

Appendix — full data tables

Every table from the raw `report.md` , so this document is self-contained. These are a snapshot; if you collect more ratings or re-score and re-run `stage5_report.py` , refresh these from the regenerated `report.md` .

A1. Exact model IDs used

model	response model id(s)
claude-haiku-4-5	claude-haiku-4-5-20251001
claude-sonnet-4-6	claude-sonnet-4-6
gemini-2.5-flash	gemini-2.5-flash
gemini-3.5-flash	gemini-3.5-flash
gpt-4o	gpt-4o-2024-08-06

A2. Consistency, per model per room (within/between std)

model	room_type	n_images	within_image_std	between_image_std	ratio_within_over_between
claude-haiku-4-5	bathroom	38	0.06	0.775	0.078
claude-haiku-4-5	bedroom	38	0.107	1.368	0.078
claude-haiku-4-5	kitchen	38	0.09	1.119	0.08
claude-haiku-4-5	living_room	38	0.126	0.938	0.135
claude-sonnet-4-6	bathroom	38	0.108	1.234	0.087
claude-sonnet-4-6	bedroom	38	0.105	1.367	0.077
claude-sonnet-4-6	kitchen	38	0.092	1.465	0.063
claude-sonnet-4-6	living_room	38	0.103	1.333	0.077
gemini-2.5-flash	bathroom	38	0.096	1.034	0.093
gemini-2.5-flash	bedroom	38	0.166	1.291	0.129
gemini-2.5-flash	kitchen	38	0.101	1.495	0.067
gemini-2.5-flash	living_room	38	0.113	1.209	0.093
gemini-3.5-flash	bathroom	38	0.043	1.076	0.04
gemini-3.5-flash	bedroom	38	0.058	1.133	0.052
gemini-3.5-flash	kitchen	38	0.044	1.318	0.033
gemini-3.5-flash	living_room	38	0.049	1.286	0.038
gpt-4o	bathroom	38	0.102	1.086	0.094
gpt-4o	bedroom	38	0.079	1.416	0.056
gpt-4o	kitchen	38	0.12	1.425	0.084
gpt-4o	living_room	38	0.129	1.179	0.11

A3. Scale-compression flags, per model per room

model	room_type	share_in_extremes	range_used	scale_span	compressed
claude-haiku-4-5	bathroom	0	4	7	True
claude-haiku-4-5	bedroom	0.105	4.5	7	False
claude-haiku-4-5	kitchen	0.068	3.7	6	False
claude-haiku-4-5	living_room	0.026	4.5	7	True
claude-sonnet-4-6	bathroom	0	4.3	7	True
claude-sonnet-4-6	bedroom	0.016	4.8	7	True
claude-sonnet-4-6	kitchen	0.258	4.6	6	False
claude-sonnet-4-6	living_room	0.026	5.5	7	True
gemini-2.5-flash	bathroom	0	3.8	7	True
gemini-2.5-flash	bedroom	0	4.7	7	True
gemini-2.5-flash	kitchen	0.311	4.8	6	False
gemini-2.5-flash	living_room	0.026	4.9	7	True
gemini-3.5-flash	bathroom	0	4	7	True
gemini-3.5-flash	bedroom	0	4.5	7	True
gemini-3.5-flash	kitchen	0.2	4.5	6	False
gemini-3.5-flash	living_room	0.026	5.1	7	True
gpt-4o	bathroom	0	3.5	7	True
gpt-4o	bedroom	0.137	5	7	False
gpt-4o	kitchen	0.411	4.8	6	False
gpt-4o	living_room	0.026	5.3	7	True

A4. Spearman vs gpt-4o, per room

model	overall	kitchen	bathroom	bedroom	living_room
claude-haiku-4-5	0.818	0.838	0.804	0.903	0.823
claude-sonnet-4-6	0.921	0.956	0.94	0.932	0.879
gemini-2.5-flash	0.926	0.946	0.929	0.884	0.861
gemini-3.5-flash	0.938	0.968	0.964	0.922	0.91

Disagreements vs gpt-4o (|mean diff| ≥ 2 levels): **None.**

A5. Room-type disagreement rate (all zero)

Every model, every room type: `disagreement_rate = 0` . The scoring-call room judgment never contradicted the Stage 1 label.

A6. Cost (full)

model	mean_cost_per_call_usd	mean_latency_s	est_scoreable_images	full_run_standard_usd	full_run_batch_usd
claude-haiku-4-5	0.0046	4.506	41779	193.891	96.9453
claude-sonnet-4-6	0.0058	8.9176	41779	243.095	121.548
gemini-2.5-flash	0.003	8.204	41779	125.385	62.6925
gemini-3.5-flash	0.0105	7.319	41779	439.455	219.727
gpt-4o	0.0049	5.8076	41779	202.851	101.425

A7. Human references

reference	kind	raters	n_images
all	group	harvey, robin, seb	152
harvey_robin	group	harvey, robin	152
harvey	individual	harvey	152
robin	individual	robin	152
seb	individual	seb	152

A8. Model vs human — Spearman rho (full matrix)

model	all	harvey_robin	harvey	robin	seb
gpt-4o	0.742	0.723	0.736	0.611	0.62
gemini-3.5-flash	0.795	0.776	0.763	0.673	0.66
gemini-2.5-flash	0.725	0.703	0.708	0.596	0.616
claude-haiku-4-5	0.638	0.624	0.606	0.547	0.531
claude-sonnet-4-6	0.758	0.739	0.731	0.639	0.637

A9. Model vs human — mean absolute deviation (full matrix)

model	all	harvey_robin	harvey	robin	seb
gpt-4o	0.896	0.851	0.881	1.067	1.223
gemini-3.5-flash	0.672	0.693	0.752	0.911	1.003
gemini-2.5-flash	0.792	0.843	0.847	1.029	1.052
claude-haiku-4-5	0.813	0.827	0.877	1.053	1.082
claude-sonnet-4-6	0.786	0.783	0.827	0.976	1.092

A10. Model vs human — signed bias, model minus human (full matrix)

model	all	harvey_robin	harvey	robin	seb
gpt-4o	0.684	0.469	0.371	0.568	1.114
gemini-3.5-flash	0.446	0.232	0.133	0.33	0.876
gemini-2.5-flash	0.42	0.205	0.106	0.304	0.85
claude-haiku-4-5	0.452	0.237	0.138	0.336	0.882
claude-sonnet-4-6	0.484	0.269	0.17	0.368	0.914

A11. Human vs human — inter-rater matrix and per-rater agreement

	harvey	robin	seb
harvey	1	0.68	0.672
robin	0.68	1	0.589
seb	0.672	0.589	1

Human-agreement ceiling (mean pairwise rho): **0.647**.

rater	n_rated	mean_rho_vs_others	outlier
harvey	152	0.676	False
robin	152	0.635	False
seb	152	0.63	False

A12. Recommendation candidates

model	spearman_vs_human_mean	consistency_ratio	mean_cost_per_call_usd
gemini-3.5-flash	0.795	0.0407	0.0105
claude-sonnet-4-6	0.7585	0.0761	0.0058
gpt-4o	0.7424	0.0859	0.0049
claude-haiku-4-5	0.638	0.0928	0.0046
gemini-2.5-flash	0.7254	0.0955	0.003

Pick: **gemini-3.5-flash**.

A13. Per-room human ceiling (pairwise Spearman among harvey/robin/seb)

This is how much the humans agree *with each other* within each room — the practical upper bound on what any model can hit. Bedroom and bathroom are the noisiest (humans disagree most); kitchen and living room are the steadiest.

room	ceiling	harvey-robin	harvey-seb	robin-seb
kitchen	0.719	0.81	0.63	0.72
bathroom	0.626	0.68	0.66	0.54
bedroom	0.598	0.57	0.71	0.51
living_room	0.707	0.74	0.70	0.68

A14. Model vs human, per room type (the prompt-tuning view)

Everything in section 7 (rho / MAD / bias) but split by room, against two references: harvey+robin (2 raters) and all (all 3). ~38 images per room. Read this against the A13 ceilings: where rho already meets or beats the ceiling, ranking is saturated and the only fixable error is bias (level offset).

rho — do model and human rank the room the same? (higher = better)

Reference: harvey+robin

model	kitchen	bathroom	bedroom	living_room
gpt-4o	0.846	0.778	0.587	0.814
gemini-3.5-flash	0.862	0.769	0.710	0.780
gemini-2.5-flash	0.863	0.703	0.576	0.731
claude-haiku-4-5	0.762	0.532	0.575	0.604
claude-sonnet-4-6	0.887	0.746	0.623	0.694

Reference: all (h+r+seb)

model	kitchen	bathroom	bedroom	living_room
gpt-4o	0.837	0.769	0.657	0.830
gemini-3.5-flash	0.850	0.748	0.784	0.810
gemini-2.5-flash	0.844	0.698	0.697	0.731
claude-haiku-4-5	0.747	0.512	0.632	0.643
claude-sonnet-4-6	0.867	0.733	0.685	0.727

MAD — how many points off, typically? (lower = better)

Reference: harvey+robin

model	kitchen	bathroom	bedroom	living_room
gpt-4o	1.034	0.583	0.942	0.844
gemini-3.5-flash	0.682	0.649	0.695	0.747
gemini-2.5-flash	0.872	0.762	0.900	0.841
claude-haiku-4-5	0.667	0.801	0.894	0.946
claude-sonnet-4-6	0.751	0.687	0.893	0.799

Reference: all (h+r+seb)

model	kitchen	bathroom	bedroom	living_room
gpt-4o	1.163	0.617	0.889	0.913
gemini-3.5-flash	0.762	0.599	0.643	0.683
gemini-2.5-flash	0.975	0.618	0.768	0.806
claude-haiku-4-5	0.718	0.838	0.846	0.852
claude-sonnet-4-6	0.884	0.665	0.870	0.724

Signed bias — model minus human (positive = model scores high)

Reference: harvey+robin

model	kitchen	bathroom	bedroom	living_room
gpt-4o	0.901	0.029	0.324	0.623
gemini-3.5-flash	0.459	-0.159	0.263	0.363
gemini-2.5-flash	0.621	-0.308	0.107	0.399
claude-haiku-4-5	0.340	0.265	0.049	0.294
claude-sonnet-4-6	0.496	-0.045	0.343	0.282

Reference: all (h+r+seb)

model	kitchen	bathroom	bedroom	living_room
gpt-4o	1.045	0.314	0.526	0.851
gemini-3.5-flash	0.604	0.126	0.465	0.591
gemini-2.5-flash	0.766	-0.023	0.309	0.628
claude-haiku-4-5	0.485	0.550	0.251	0.522
claude-sonnet-4-6	0.641	0.240	0.545	0.510

A15. Mean score levels per room (why the bias is positive)

Every model's average sits **above** the human average in every room; the gap is biggest in kitchen and living room. This is the calibration target — pull the model's typical level down toward the human mean, especially in those two rooms.

source	kitchen	bathroom	bedroom	living_room
humans (all 3)	4.00	4.28	4.06	4.25
gpt-4o	5.05	4.59	4.59	5.10
gemini-3.5-flash	4.60	4.41	4.53	4.84
gemini-2.5-flash	4.77	4.26	4.37	4.87
claude-haiku-4-5	4.48	4.83	4.31	4.77
claude-sonnet-4-6	4.64	4.52	4.61	4.76

Takeaway for prompt tuning (all models show the same pattern → it's the prompt/anchors, not the model):

ranking is already at or above the human ceiling in every room, so there's no headroom left there. The remaining fixable error is the upward level bias, concentrated in **kitchen** and **living room**. Bathroom is the best-calibrated room (bias ≈ 0) — treat it as the template. Bedroom has the lowest human ceiling (0.598), so its noise is largely irreducible; don't over-invest there.

A16. Share of images where gemini-3.5-flash is off by more than 1 point

Per-image gap between gemini's mean score (5 replicates) and the human mean. 38 images per room. "% >1" is the headline; the last columns count the larger misses. Almost all of these are gemini scoring *high*.

Reference: all (h+r+seb)

room	n	images >1 off	% >1	# >1.5	# >2
kitchen	38	13	34.2%	4	1
living_room	38	10	26.3%	4	2
bedroom	38	7	18.4%	3	1
bathroom	38	6	15.8%	1	0
all	152	36	23.7%	12	4

Reference: harvey+robin

room	n	images >1 off	% >1	# >1.5	# >2
kitchen	38	9	23.7%	2	1
bedroom	38	11	28.9%	3	1
living_room	38	10	26.3%	4	2
bathroom	38	6	15.8%	1	0
all	152	36	23.7%	10	4

Same ranking as the bias/MAD tables: kitchen worst, bathroom best. "More than 1 point" is a loose bar relative to the human ceiling — most of these are 1–1.5 off, and only 4 of 152 exceed 2 points. A visual gallery of every >1 image (photo + each rater's score + gemini's score) is generated separately by `make_diff_gallery.py`.

A17. Prompt calibration pass (baseline → v1 → v2)

After the analysis above showed gemini-3.5-flash ranking at/above the human ceiling but scoring systematically **high** (worst in kitchen and living room), we edited the per-room scoring prompts to fix the *level* without disturbing the *ranking*, and re-scored the same 152 images (gemini only, 1 call/image, ~\$1.75 per pass). Two iterations:

- **v1** — strengthened "ignore staging," reset the "standard renovation = Level 4" baseline, and added per-room caps. Result: kitchen and living bias went to ~0, **but** the caps over-compressed the scale — the top of the range collapsed (share ≥ 5.5 fell to ~5% in every room), bathroom over-corrected to -0.61, and rank correlation dropped (living rho 0.81→0.68).
- **v2** — walked bathroom back to near-original, softened the numeric caps to "furniture/styling alone don't raise the score," and added an explicit "do not compress to the middle — premium permanent finishes still deserve 6–7" rule.

Signed bias (model – human), all-3 reference — baseline → v1 → v2 (target 0):

room	baseline	v1	v2
kitchen	+0.60	-0.05	+0.20
bathroom	+0.13	-0.61	-0.08
bedroom	+0.47	+0.27	+0.34
living_room	+0.59	+0.03	+0.17
mean bias 	0.45	0.24	0.20

Spearman rho vs human, all-3 — baseline → v1 → v2 (higher = better; the last column is the per-room human ceiling — the model beats it in every room at v2):

room	baseline	v1	v2	human ceiling
kitchen	0.850	0.824	0.822	0.719
bathroom	0.748	0.754	0.724	0.626
bedroom	0.784	0.721	0.719	0.598
living_room	0.810	0.681	0.772	0.707
mean	0.798	0.745	0.759	

Scale spread — share of images scored ≥ 5.5 and score std (baseline $\rightarrow$ v1 $\rightarrow$ v2). v1 collapsed the top of the scale; v2 restored it:

room	share ≥ 5.5 (b $\rightarrow$ v1 $\rightarrow$ v2)	std (b $\rightarrow$ v1 $\rightarrow$ v2)
kitchen	32% $\rightarrow$ 5% $\rightarrow$ 16%	1.32 $\rightarrow$ 1.00 $\rightarrow$ 1.11
bathroom	16% $\rightarrow$ 5% $\rightarrow$ 11%	1.08 $\rightarrow$ 0.90 $\rightarrow$ 1.01
bedroom	16% $\rightarrow$ 5% $\rightarrow$ 11%	1.13 $\rightarrow$ 0.79 $\rightarrow$ 0.86
living_room	37% $\rightarrow$ 13% $\rightarrow$ 21%	1.29 $\rightarrow$ 1.05 $\rightarrow$ 1.07

Verdict: v2 is the chosen prompt. Vs baseline it cuts mean absolute bias **55% (0.45 $\rightarrow$ 0.20)** with every room within ± 0.35 , restores the full scale range, and keeps rank correlation **above the human ceiling in all four rooms** (so the nominal rho dip vs baseline is not practically meaningful). Residual: bedroom still runs $\sim +0.34$ high, but bedroom has the lowest human ceiling and is measured here on a mid-heavy 38-image sample with single (non-replicated) calls, so further prompt tuning risks overfitting. Next steps are the anchor-grid images (unchanged so far) and validation on a price-stratified sample that covers the scale tails, before the full-corpus production run.

Self-contained companion to [report.md](#). To turn this into a PDF: `pandoc report_explained.md -o report_explained.pdf` (or open it in any Markdown viewer and print to PDF).